1 Setting

The hypotheses of Solomonoff axiom induction are axiom systems, sets of first-order sentences produced by programs; the word "axiom" refers to them, not to the induction axiom. The data are sentences labelled true or false. A hypothesis is compatible with the data when it is consistent and proves every datum: it must *explain* each datum, not merely tolerate it. Randomness enters as in the one-tape version of Solomonoff operator induction (Hänni and Claude (Anthropic), 2026b), which we call the companion paper: a universal machine reads one random tape and emits sentences, and the prior probability of a data set is the probability that the emitted theory is consistent and proves every datum. This section fixes the definitions; **??** develops their basic theory, and the proofs are in **??**.

1.1 Language, background and data

Definition 1.1 (Setting and data)**.** $\mathcal{L}$ is a countable first-order language with equality and computable syntax, $\mathrm{Sent} := \mathrm{Sent}_{\mathcal{L}}$, and $\vdash$ is derivability in a fixed sound and complete calculus. The *background* $B \subseteq \mathrm{Sent}$ is decidable and consistent; by default $B = \emptyset$. Put $\mathrm{Cn}(\Gamma) := \{\varphi : B \cup \Gamma \vdash \varphi\}$. "$\Gamma$ proves φ" means $B \cup \Gamma \vdash \varphi$; Γ is *consistent* if $B \cup \Gamma$ is; Γ *decides* φ if it proves φ or $\neg\varphi$; $\varphi^1 := \varphi$ and $\varphi^0 := \neg\varphi$.

A *data set* is a finite partial function $D : \mathrm{Sent} \rightharpoonup \{0,1\}$ (label 1 means true); a *three-valued* data set maps into $Y_3 := \{1, 0, \iota\}$, where ι means "independent". Put $D^{\pm} := \{\varphi^{D(\varphi)} : D(\varphi) \in \{0,1\}\}$, and let ψ_D be the conjunction of $D^{\pm}$ in a fixed canonical order ($\top$ if $D^{\pm} = \emptyset$); D is *consistent* if $B \cup D^{\pm}$ is. $D + (\varphi, y)$ is defined for $\varphi \notin \operatorname{dom} D$. A *reveal sequence* $(\varphi_t, y_t)_{t \le n}$ is consistent if $D_n := \{(\varphi_t, y_t) : t \le n\}$ is a partial function; D_t are its prefixes. An infinite stream D_∞ has $D_\infty^{\pm} := \bigcup_n D_n^{\pm}$. For a map ν from data sets to $[0,1]$ the *conditional* is $\nu(y \mid D, \varphi) := \nu(D + (\varphi, y))/\nu(D)$ for $\varphi \notin \operatorname{dom} D$ (0 if $\nu(D) = 0$), and $\nu(y \mid D, \varphi) := \mathbf{1}[y = D(\varphi)]$ for $\varphi \in \operatorname{dom} D$ (the repeated-input convention of the companion paper).

An r.e. background would serve as well, since by Craig's trick it has a decidable axiomatisation with the same consequences. The data $(\varphi, 1)$ and $(\neg\varphi, 0)$ are different data with the same content; weights will depend only on content (**??**(c)).

Algorithmic notation. V is a fixed universal prefix machine, K prefix complexity, $\mathbf{m}(i) := \sum_{V(p)=i} 2^{-|p|} \ge 2^{-K(i)}$ and $\Omega_V := \sum_{p \in \operatorname{dom} V} 2^{-|p|}$; W_e is the e-th r.e. set of sentences. $\mathbb{U}$ is the fair-coin measure on tapes $\omega \in \{0,1\}^\infty$, $[w]$ the cylinder of a finite string w, and $\omega_{1:n}$ the first n bits of ω. Logarithms are binary and losses are in bits. The constants $c, c', \ldots$ depend only on the fixed universal machines, the coding of syntax and B; "lsc" means lower semicomputable. We use two standard facts (see Li and Vitányi, 2008): $\mathbf{m}$ dominates every lsc discrete semimeasure up to a constant factor; and the conditional coding theorem: if $P(a \mid z)$ is lsc uniformly in (a, z) and $\sum_a P(a \mid z) \le 1$, then $K(a \mid z) \le -\log P(a \mid z) + K(P) + c$.

Definition 1.2 (Literal setting; fresh atoms)**.** In the *literal setting*, $\mathcal{L}$ contains a unary relation symbol R that does not occur in B, and closed terms t_w ($w \in \{0,1\}^*$) that some model of B interprets as pairwise distinct elements. *Fresh atoms* are sentences $R(t_w)$ for pairwise distinct words w; so $R(t_w)$ and $\neg R(t_w)$ are never two fresh atoms. The standard instance is the *literal language*: a constant $\mathtt{e}$, unary function symbols $\mathtt{s}_0, \mathtt{s}_1$ and R, with $t_w := \mathtt{s}_{w_1}(\cdots \mathtt{s}_{w_k}(\mathtt{e}) \cdots)$ and $\varphi_w := R(t_w)$, of $|w| + 2$ symbols; its *named structure* $\mathfrak{T}$ is the free term algebra on $\mathtt{e}, \mathtt{s}_0, \mathtt{s}_1$. For a set of words $X \subseteq \{0,1\}^*$, D^X lists $(\varphi_w, \mathbf{1}[w \in X])$ in length-lexicographic order and f_X is the

literal labeller of X. The *fit class* of an inducer is the set of X on whose sequence D^X it has bounded loss.

Every result stated for fresh atoms or on literal data assumes this setting. Two facts carry all such uses (**??**): a set of literals of R with at most one literal per word is consistent with B, and a consistent such set proves exactly its own members among the literals.

1.2 Theory machines and their laws

A hypothesis of Solomonoff axiom induction is produced by a machine that reads random bits and writes sentences. A function machine of Solomonoff operator induction must give each input at most one label; a theory machine may write anything. Emitting more axioms is always allowed, and consistency is a property of the output, a random event.

Definition 1.3 (Theory machine)**.** A *theory machine* is a c.e. set $G \subseteq \{0,1\}^* \times \mathrm{Sent}$ ("after reading w from the tape, emit χ"). For finite w put $A^G_w := \{\chi : (v,\chi) \in G,\ v \sqsubseteq w\}$, and for a tape ω let $A^G_\omega := \bigcup_n A^G_{\omega_{1:n}}$ be the axioms emitted on ω. Let

$$O^G_\chi := \{\omega : B \cup A^G_\omega \vdash \chi\}, \qquad \mathrm{Cons}^G := \{0,1\}^\infty \setminus O^G_\bot, \qquad E^G_D := O^G_{\psi_D} \cap \mathrm{Cons}^G.$$

The *law* of G is $\nu_G(D) := \mathbb{U}(E^G_D)$, the probability that the random theory is consistent and proves every datum. The *three-valued law* is $\nu^3_G(D) := \mathbb{U}(E^{3,G}_D)$ with $E^{3,G}_D := E^G_D \cap \bigcap_{D(\varphi)=\iota} (\{0,1\}^\infty \setminus (O^G_\varphi \cup O^G_{\neg\varphi}))$: in addition, the theory decides no ι-labelled sentence. On Cons^G the *trichotomy function* $\mathrm{tri}^G_\omega : \mathrm{Sent} \to Y_3$ is 1 at the sentences $B \cup A^G_\omega$ proves, 0 at those it refutes and ι elsewhere. (G_i) is a fixed effective enumeration of the c.e. subsets of $\{0,1\}^* \times \mathrm{Sent}$; $K(G) := \min\{K(i) : G_i = G\}$ and, for a law ν, $K(\nu) := \min\{K(i) : \nu_{G_i} = \nu\}$.

These events are measurable: O^G_χ is effectively open uniformly in χ and an index of G, Cons^G is a Π^0_1 class, and $\nu_G(D) = \mathbb{U}(O^G_{\psi_D}) - \mathbb{U}(O^G_\bot)$ (**??**). *Examples.* (i) A deterministic r.e. theory W_e: $G = \{(\epsilon,\chi) : \chi \in \mathsf{W}_e\}$, with $\nu_G(D) = \mathbf{1}[B \cup \mathsf{W}_e \text{ consistent and } \vdash \psi_D]$. (ii) Fair-coin facts on a family $(\alpha_j)_{j\geq 1}$ of sentences: $G = \{(w, \alpha_j^{w_j}) : |w| = j\}$. (iii) A theory plus coin facts: the union of (i) and (ii). (iv) A finite mixture of finite theories $T_1,\dots,T_r$: $G = \{(w_j,\chi) : \chi \in T_j\}$ for a prefix-free set $\{w_1,\dots,w_r\}$, which gives T_j weight $2^{-|w_j|}$; a tape with no prefix among the w_j emits nothing, and its theory is the empty one, consistent since B is. More generally, a theory machine can complete a law at random inside what the law allows; such randomised theories are how approximate laws enter Solomonoff axiom induction (**??**).

Definition 1.4 (Solomonoff axiom induction)**.** The *universal theory machine* is $G_U := \{(pw,\chi) : p \in \mathrm{dom}\, V,\ (w,\chi) \in G_{V(p)}\}$: it reads a self-delimiting program and runs that machine on the rest of the tape. Write $A_\omega := A^{G_U}_\omega$, $O_\chi := O^{G_U}_\chi$, $\mathrm{Cons} := \mathrm{Cons}^{G_U}$, $E_D := E^{G_U}_D$, $E^3_D := E^{3,G_U}_D$ and $\mathrm{tri}_\omega := \mathrm{tri}^{G_U}_\omega$.

- *SAI* is the one-tape proof-requiring law $M_A := \nu_{G_U}$. Put $M^3_A := \nu^3_{G_U}$, $L_A := -\log M_A$ and $\mathrm{Inc}_0 := \mathbb{U}(O_\bot)$, the probability that the random theory is inconsistent.
- The *posterior* after D with $M_A(D) > 0$ is the law of A_ω under $\mathbb{U}(\cdot \mid E_D)$; three-valued, under $\mathbb{U}(\cdot \mid E^3_D)$.
- *Trichotomy.* $\mathrm{Bel}_D(\chi) := \mathbb{U}(E_D \cap O_\chi)/\mathbb{U}(E_D)$ is the posterior probability that the random theory proves χ; $\mathrm{Dis}_D(\chi) := \mathrm{Bel}_D(\neg\chi)$; $\mathrm{Indep}_D(\chi) := 1 - \mathrm{Bel}_D(\chi) - \mathrm{Dis}_D(\chi)$.

- The *prior probability of provability* $\mathrm{Bel}_0(\chi) := M_A(\{(\chi, 1)\}) = \mathbb{U}(O_\chi) - \mathrm{Inc}_0$ is not normalised: $\mathrm{Bel}_0(\top) = 1 - \mathrm{Inc}_0$. Its normalisation is $\widehat{\mathrm{Bel}}_0 := \mathrm{Bel}_0/(1 - \mathrm{Inc}_0) = \mathrm{Bel}_\emptyset$. The *plausibility* is $\mathrm{Pl}_0(\chi) := \mathrm{Bel}_0(\top) - \mathrm{Bel}_0(\neg\chi)$, and $\mathrm{Pl}_D(\chi) := 1 - \mathrm{Bel}_D(\neg\chi)$.
- *Class probabilities.* For a class $\mathcal{C}$ of deductively closed theories, $\mathbb{U}_A(\mathcal{C}; D) := \mathbb{U}\{\omega : B \cup A_\omega \text{ consistent}, \vdash \psi_D, \mathrm{Cn}(A_\omega) \in \mathcal{C}\}$ and $\mathbb{U}_A(\mathcal{C}) := \mathbb{U}_A(\mathcal{C}; \emptyset)$. For a theory T, $\mathbb{U}_A[T] := \mathbb{U}\{\omega : B \cup A_\omega \text{ consistent}, \mathrm{Cn}(A_\omega) = \mathrm{Cn}(T)\}$ and $\mathbb{U}_A(\mathrm{Ext}T) := \mathbb{U}\{\omega : B \cup A_\omega \text{ consistent}, \mathrm{Cn}(A_\omega) \supseteq \mathrm{Cn}(T)\}$.

If $\omega = p\omega'$ with $p \in \mathrm{dom}\, V$, then $A_\omega = A_{\omega'}^{G_{V(p)}}$, since $\mathrm{dom}\, V$ is prefix-free; a tape with no prefix in $\mathrm{dom}\, V$ emits nothing. So G_U reads one tape in two parts, a program and that program's coins. By **??**, $M_A(D) = \mathrm{Bel}_0(\psi_D)$, $M_A(b \mid D, \varphi) = \mathrm{Bel}_D(\varphi^b)$ and $M_A^3(\iota \mid D, \varphi) = \mathrm{Indep}_D(\varphi)$ for $\varphi \notin \mathrm{dom}\, D$: the predictions of SAI are posterior probabilities that the random theory proves, refutes, or leaves open the queried sentence.

1.3 Conventions and the two defects

Definition 1.5 (Loss conventions)**.** • *Semimeasure convention* (primary): the cumulative loss of a consistent reveal sequence is $\sum_{t \le n} -\log M_A(y_t \mid D_{t-1}, \varphi_t)$, which equals $L_A(D_n) - L_A(\emptyset)$ (**??**(d)); for the coin-free version it is $-\log W_A(D_n) + \log W_A(\emptyset)$.

- *Normalised convention*: each conditional is divided by $\mathrm{Bel}_D(\varphi) + \mathrm{Dis}_D(\varphi)$.
- *Three-valued convention*: labels in Y_3, scored by M_A^3, whose conditionals sum to 1 (**??**(f)).
- The consistency-requiring predictor (Definition 1.10) has its own loss, $\sum_t -\log \mathrm{Bel}^{\mathrm{nc}}_{D_{t-1}}(\varphi_t^{b_t})$.

Upper bounds proved in the semimeasure convention hold in the normalised convention, since each normalised conditional is at least the raw one. Lower bounds do not transfer: a lower bound in the semimeasure convention can fail in the normalised convention by an unbounded amount (**??**), and the normalised loss depends on the reveal order (**??**). Every lower bound below names its convention; statements about K_A, K_{fin} and the posterior need none.

The missing mass of M_A has two sources, and we keep both rather than renormalise.[1]

Remark 1.6 (Two defects). M_A has a *global* defect $1 - M_A(\emptyset) = \mathrm{Inc}_0 > 0$ and a *local* defect $\mathrm{Indep}_D(\varphi)$ at each query (**??**(b), (e)). The one-tape law M_F of SOI has only the local one ($M_F(\emptyset) = 1$), because function machines are consistent by definition. The correspondence is: "proves φ" is a Σ_1 property of the tape, like "halts with output 1"; "independent" is Π_1, like "diverges"; "inconsistent" corresponds to "halts with both outputs", which a function machine cannot do. Removing the independent mass and renormalising, or splitting it equally between the two labels, gives incoherent predictions (Examples 2.15, 2.16 and Proposition 2.17 of Hänni and Claude (Anthropic) (2026a)); keeping it, the pair $(\mathrm{Bel}_D, \mathrm{Pl}_D)$ is a coherent lower and upper probability (**??**).

1.4 Versions and complexities of theories

As for SOI, the one-tape law has a coin-free version, a mixture over deterministic programs, and a Kolmogorov version, the shortest compatible program.

[1] The proof-requiring and consistency-requiring variants, the three outcomes true, false and independent, and the choice to keep the defect of the semimeasure rather than renormalise first appeared in Hänni's working notes (Hänni, 2026).

Definition 1.7 (Versions)**.** For a data set D:

- $W_A(D) := \sum_{p\in\operatorname{dom} V} 2^{-|p|}\mathbf{1}[B\cup\mathsf{W}_{V(p)}$ consistent and $\vdash\psi_D] = \sum_e \mathbf{m}(e)\mathbf{1}[B\cup\mathsf{W}_e$ consistent and $\vdash\psi_D]$ (coin-free, over enumerators);
- $K_A(D) := \min\{|p| : p\in\operatorname{dom} V,\ B\cup\mathsf{W}_{V(p)}$ consistent and $\vdash\psi_D\}$ (∞ if there is none; the Kolmogorov version);
- W_A^{dec}, K_A^{dec}: the same over *deciders*, the $p\in\operatorname{dom} V$ such that $V(p)$ is an index of a total $\{0,1\}$-valued function d, read as the axiom set $\{\chi : d(\chi)=1\}$ (Definition 6.2 of Hänni and Claude (Anthropic) (2026a));
- $W_A^{\mathrm{gen},t}$: the same over enumerators of the delay form (Definition E.10 of Hänni and Claude (Anthropic) (2026a)), which write each axiom a within $O(t(|a|))$ steps of the previous one;
- $K_{\mathrm{fin}}(D) := \min\{K(\psi) : \psi\in\mathrm{Sent},\ B\cup\{\psi\}$ consistent, $B\cup\{\psi\}\vdash\psi_D\}$ and $W_{\mathrm{fin}}(D) := \sum_\psi \mathbf{m}(\psi)\mathbf{1}[B\cup\{\psi\}$ consistent, $B\cup\{\psi\}\vdash\psi_D]$ (finite axiomatisations).

Remark 1.8 (Deciders, enumerators and polynomial-delay enumerators agree)*.* There is c such that for every D: (a) $2^{-c}W_A^{\mathrm{dec}}(D)\le W_A(D)\le 2^cW_A^{\mathrm{dec}}(D)$ and $|K_A(D)-K_A^{\mathrm{dec}}(D)|\le c$; (b) under **??**, for t time-constructible, nondecreasing and superadditive with $t(m)\ge m$ and $t(m)\ge m^{a^*}$ (a^* fixed by the universal machine): $2^{-c}W_A(D)\le W_A^{\mathrm{gen},t}(D)\le 2^cW_A(D)$, and likewise for the Kolmogorov versions; (c) $K_A(D)\le K_{\mathrm{fin}}(D)+c\le K(\psi_D)+c$ if D is consistent; if D is inconsistent, $K_A(D)=K_{\mathrm{fin}}(D)=\infty$, while $K(\psi_D)$ is finite.

Part (a) is Craig's trick between two kinds of axiom hypotheses; (b) uses the padded Craig sets of Hänni and Claude (Anthropic) (2026a, Lemma E.12), proved there under the self-simulation assumption, which we do not re-derive (**??**). The inequality $K_A\le K_{\mathrm{fin}}+c$ can be far from tight: $K_{\mathrm{fin}}-K_A$ is unbounded (**??**).

Definition 1.9 (Complexities of theories)**.** For an r.e. theory T: $[T] := \{e : \mathrm{Cn}(\mathsf{W}_e)=\mathrm{Cn}(T)\}$; $K_{\equiv}(T) := \min_{e\in[T]} K(e)$, the complexity of T up to deductive equivalence; $\mathbf{m}_{\equiv}(T) := \sum_{e\in[T]}\mathbf{m}(e)$; $\mathrm{Ext}(T) := \{e : B\cup\mathsf{W}_e$ consistent, $\mathrm{Cn}(\mathsf{W}_e)\supseteq\mathrm{Cn}(T)\}$ and $\mathbf{m}(\mathrm{Ext}T) := \sum_{e\in\mathrm{Ext}T}\mathbf{m}(e)$. If $B\cup T$ is consistent, then $[T]\subseteq\mathrm{Ext}T$, so $2^{-K_{\equiv}(T)}\le\mathbf{m}_{\equiv}(T)\le\mathbf{m}(\mathrm{Ext}T)$, and $\mathbb{U}_A[T]\ge 2^{-c}\mathbf{m}_{\equiv}(T)$, $\mathbb{U}_A(\mathrm{Ext}T)\ge 2^{-c}\mathbf{m}(\mathrm{Ext}T)$ (**??**(i)). The hypothesis is needed: for $T=\{\bot\}$, $\mathbb{U}_A[T]=0$ and $\mathrm{Ext}T=\emptyset$, while $\mathbf{m}_{\equiv}(T)\ge 2^{-K_{\equiv}(T)}>0$.

1.5 Variants and questions

Proof-requiring compatibility asks the hypothesis to prove each datum. Two weaker requirements are natural: that the hypothesis be consistent with the data, or that it be graded by how many data it proves. A third variant replaces labelled data by sentences sampled from the theory. **??** explains why we take the proof-requiring law as primary.

Definition 1.10 (Variants)**.** Let $\varepsilon_\emptyset := \mathbb{U}\{\omega : A_\omega=\emptyset\}$, the mass of the empty theory; $\varepsilon_\emptyset\ge 2^{-|p_\emptyset|}>0$ for a V-program $p_\emptyset$ of an index of the empty machine.

- *Consistency-requiring,* 0/1: $M_A^{\mathrm{nc}}(D) := \mathbb{U}\{\omega : B\cup A_\omega\cup D^{\pm}$ consistent$\}$. Its posterior is the law of A_ω given that event, and its predictor uses the hypothesis together with the data: $\mathrm{Bel}_D^{\mathrm{nc}}(\varphi) := \mathbb{U}\{\omega : B\cup A_\omega\cup D^{\pm}$ consistent and $\vdash\varphi\}/M_A^{\mathrm{nc}}(D)$ and $\mathrm{Dis}_D^{\mathrm{nc}}(\varphi) := \mathrm{Bel}_D^{\mathrm{nc}}(\neg\varphi)$.
- *Graded consistency-requiring, grade* $\frac12$: for consistent A, $\nu_A^{1/2}(D) := \mathbf{1}[B\cup A\cup D^{\pm}$ consistent$]\prod_{\varphi\in\operatorname{dom} D}\mathrm{gr}_A(\varphi)$, with $\mathrm{gr}_A(\varphi) := 1$ if $B\cup A\vdash\varphi^{D(\varphi)}$ and $\frac12$ otherwise; $M_A^{1/2}(D) := \int_{\mathrm{Cons}}\nu_{A_\omega}^{1/2}(D)\,d\mathbb{U}(\omega)$.

- *Generative*: a theory A has a law P_A on sentences, given by a derivation grammar with citation weights on the axioms (the likelihood L_1 of Definition 2.9 of Hänni and Claude (Anthropic) (2026a)); the data are unlabelled sentences $X_1, X_2, \ldots$, and the universal law of data streams is $\Xi(X_{1:n}) := \int \prod_{i \le n} P_{A_\omega}(X_i)\, d\mathbb{U}(\omega)$, with the sampling coins read from separate parts of the tape (**??**).

The graded weight is the symmetric form of the graded score S_g of Hänni and Claude (Anthropic) (2026a, Definition 2.12) with $g \equiv 1$ on proved data and $g_\infty = \frac{1}{2}$: S_g grades only the data labelled true, $\nu^{1/2}$ grades the data labelled false too. The results below use only data labelled true, where the two agree.

Definition 1.11 (Questions)**.** A *question* is a finite set $\mathcal{Q} = \{\varphi_1, \ldots, \varphi_k\}$ of pairwise B-incompatible sentences whose disjunction B proves, its *direct answers*; a yes/no question is $\{\varphi, \neg\varphi\}$. A theory *answers* $\mathcal{Q}$ if it proves some φ_i. A *conditional question* is a pair (α, ℓ), read "given α, is ℓ the case?". Under the *encoding* (E3), a theory T is compatible with the conditional datum (α, ℓ) iff $B \cup T \cup \{\alpha\}$ is consistent and proves ℓ; equivalently, $T \vdash \alpha \to \ell$ and $T \nvdash \neg\alpha$.

This follows the tradition of erotetic logic (cf. Belnap and Steel, 1976); results that need only pairwise incompatibility say so. (E3) is not a single datum of M_A^3: it combines "the label of $\alpha \to \ell$ is 1" with "the label of $\neg\alpha$ is not 1", a union of two M_A^3-events. The two-valued encoding by the datum $(\alpha \to \ell, 1)$ is degenerate, since a single world satisfying the law fits all of its question data (**??**); it is never used for conditional data. (E3) makes "given α" hypothetical: a compatible theory must leave every antecedent open.

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